



Department of Computer Science & Information Technology



Course Information

Course Code: CS-08519

Credit Hours: 4 (3+1)

Lab Hours/Week: 3

Course Title: **Computer Networks**

Lecture Hours/Week: 3

Pre-Requisites: **None**

b. Lab Learning Outcomes – Computer Networks

Students will *explain* and *apply* the fundamental concepts of networking, IP addressing, subnet masks, and default gateways. Students will troubleshoot network connectivity using commands like `ping`, `tracert`, and `netstat`, `nslookup` etc.

Students will *identify*, connect, and configure network devices, cables, and connectors using Cisco Packet Tracer.

Students will *design* and implement LAN topologies and configure switches and routers effectively.

Students will *execute* router configuration commands using the CLI and verify operational accuracy.

Students will *configure and verify* DHCP and DNS servers for automated IP assignment and name resolution.

Students will *configure* VLANs and establish intra- and inter-VLAN communication.

Students will *implement* STP and RSTP to ensure redundancy and prevent loops in switched networks.

Students will *configure and troubleshoot* static and dynamic routing protocols (RIP, OSPF, EIGRP).

Students will *configure and analyze* Frame Relay networks for WAN connectivity.

Students will *apply* Access Control Lists (ACLs) to control access and restrict specific network services.

Students will *configure and test* Static, Dynamic, and PAT-based NAT for address translation.

Students will perform Cisco IOS *backup and restoration* using TFTP for disaster recovery.

Students will integrate and evaluate complete network systems through end-to-end simulation and testing.